04 — Multimodal Text & Image Fusion

Goal: Combine state-of-the-art text (TF-IDF + SVM, CamemBERT) and image models (ConvNeXt-Tiny) to maximise classification performance.

- **Image Model**: `ConvNeXt-Tiny` (Full Fine-Tuning + Smart Cropping from `notebooks/03_image_fineTuning/03_7_image_convnext_optimized.ipynb`), reaching **0.6998 Weighted F1** alone.
- **Text Models**: Calibrated `LinearSVC` (**0.8430 Weighted F1**) and fine-tuned `CamemBERT` (**0.8765 Weighted F1**).
- **Fusion Strategies**:
  1. **Soft Voting**: Weighted probability average of the models (SVM + image, CamemBERT + image, CamemBERT + SVM + image).
  2. **Stacking Classifier (Random Forest)**: Meta-classifier trained via 5-fold cross-validation on the combined 81-dimensional probability representations (27 classes × 3 models).

1. Text probabilities

We use the optimised text model from notebook 03_2 (all features, binary TF-IDF, LinearSVC with C = 1). A LinearSVC gives scores, not probabilities. Calibration learns on the training set (3-fold CV) how to turn these scores into probabilities.

```
Text probabilities saved: (16984, 28)
Each row sums to 1      : True
```

2. Match text and image on the same products

```
Products scored by both: 16984
```

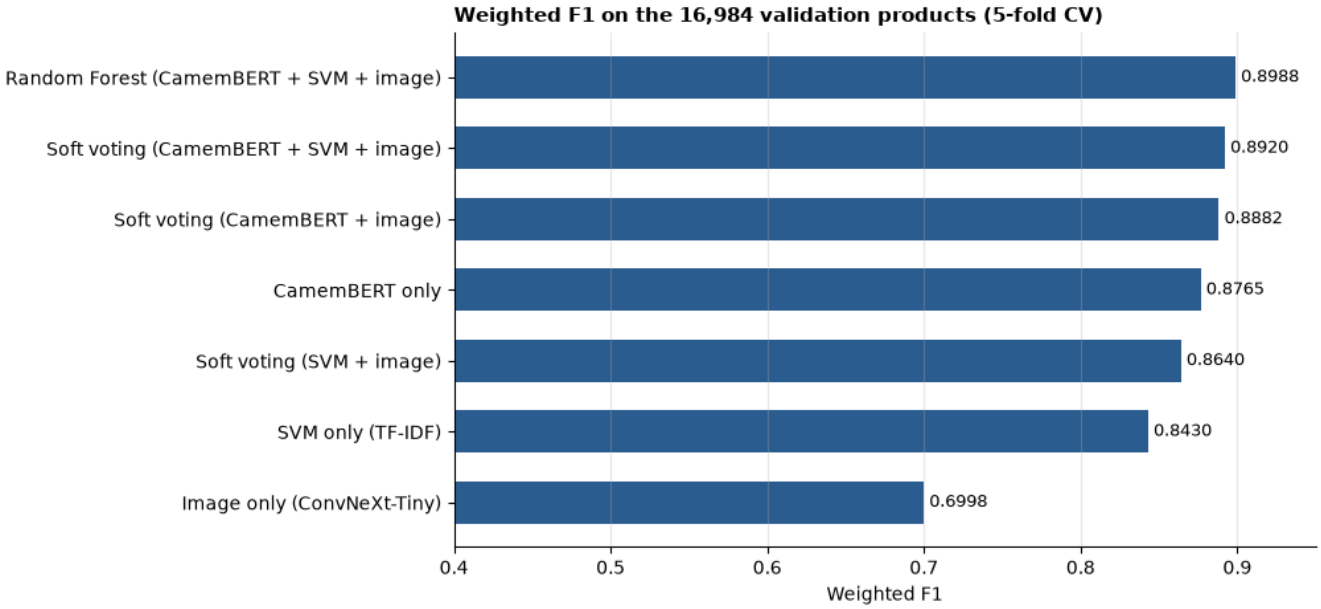
3. Combine the models

Text and image probabilities exist for the 16984 validation products, which none of the models were trained on. We score every method with 5-fold cross-validation on these products: the combiner learns on 4/5 of them and predicts the remaining 1/5. Every product is predicted by a combiner that never saw it.

Two ways to combine:

- Soft voting: a weighted average of the probabilities of each model, then the class with the highest average. The weights (steps of 0.1, summing to 1) are chosen on the 4/5 part only.
- Random Forest: a model that learns from the 81 probabilities (27 classes × 3 models) how to combine them.

```
Soft voting (SVM + image) — weights chosen in each fold: [(0.6, 0.4), (0.6, 0.4),
(0.6, 0.4), (0.6, 0.4), (0.6, 0.4)]
Soft voting (CamemBERT + image) — weights chosen in each fold: [(0.5, 0.5), (0.5,
0.5), (0.5, 0.5), (0.5, 0.5), (0.5, 0.5)]
Soft voting (CamemBERT + SVM + image) — weights chosen in each fold: [(0.4, 0.3,
0.3), (0.4, 0.3, 0.3), (0.3, 0.4, 0.3), (0.4, 0.3, 0.3), (0.4, 0.3, 0.3)]
```



Model	Weighted F1	Macro F1
Random Forest (CamemBERT + SVM + image)	0.8988	0.8856
Soft voting (CamemBERT + SVM + image)	0.8920	0.8797
Soft voting (CamemBERT + image)	0.8882	0.8730
CamemBERT only	0.8765	0.8594
Soft voting (SVM + image)	0.8640	0.8548
SVM only (TF-IDF)	0.8430	0.8326
Image only (ConvNeXt-Tiny)	0.6998	0.6752

Analysis

- The best model is the Random Forest on CamemBERT, SVM and image: 0.8988 weighted F1 and 0.8856 macro F1.
- That is +2.2 points over CamemBERT alone (0.8765) and +5.6 points over the SVM alone (0.8430).
- Soft voting on all three models (weights 0.4 CamemBERT, 0.3 SVM, 0.3 image in 4 of 5 folds) reaches 0.8920. It already gives most of the gain.
- Soft voting with CamemBERT and image (0.5 each) reaches 0.8882. With SVM and image (0.6 SVM, 0.4 image) it reaches 0.8640.
- The weights are almost the same in every fold, so the choice is stable.
- The Random Forest adds 0.7 points over the best soft voting. It can weight each model differently for each category, while soft voting uses one weight for all categories.
- We also tried Gradient Boosting instead of Random Forest (appendix notebook). It scored slightly lower and was much slower, so we kept the Random Forest.

4. Per category: best fusion vs CamemBERT

Code	Category	CamemBERT F1	Best fusion F1	Gain
2705	New Books	0.818	0.926	0.108
10	Used Books	0.689	0.788	0.099
1180	Wargames	0.614	0.706	0.093
40	Games	0.795	0.855	0.059
1302	Outdoor Toys	0.810	0.853	0.043
1140	Figurines	0.834	0.862	0.028
60	Consoles	0.876	0.903	0.027
1940	Food	0.946	0.969	0.023
1160	Cards	0.955	0.978	0.023
1320	Baby	0.856	0.877	0.021
2585	Garden Tools	0.823	0.843	0.019
2403	Book Lots	0.868	0.886	0.018
2462	Used Games	0.821	0.838	0.017
2220	Pets	0.890	0.906	0.017
2582	Garden Furniture	0.794	0.810	0.016
50	Game Accessories	0.880	0.896	0.016
1281	Board Games	0.684	0.700	0.015
2522	Stationery	0.945	0.958	0.014
2583	Pool	0.970	0.983	0.013
1920	Textiles	0.925	0.938	0.013
1280	Toys	0.791	0.803	0.012
1560	Furniture	0.882	0.890	0.008
2060	Decor	0.861	0.867	0.006
2280	Magazines	0.919	0.922	0.003
1300	RC Models	0.987	0.987	0.000
2905	PC Games	0.991	0.991	0.000
1301	Baby Socks	0.978	0.975	-0.003

Analysis

- The best fusion is better than CamemBERT on 26 of the 27 categories. Only Baby Socks drops slightly (-0.3 points).
- The largest gains are on New Books (+10.8 points), Used Books (+9.9) and Wargames (+9.3).
- Books gain the most from the image: the cover shows if a book is new, used, or a lot of several books.
- Wargames was CamemBERT's weakest category. The SVM catches its rare exact words (*warhammer*, *heroclix*), so combining both fixes it.

Conclusion

- Combining models helps. Soft voting on the three models already reaches 0.8920. A learned combination (Random Forest) adds 0.7 points more.
- Best model: Random Forest on the CamemBERT, SVM and ConvNeXt probabilities, with 0.8988 weighted F1 on the validation products (5-fold CV).
- Each part adds something: CamemBERT reads context, the SVM catches rare specific words, and the image helps on books.